2. Explain the difference between method declaration and method body.

Method declaration: Defines the method's name, return type, and parameters.

Method body: Contains the instructions or code that the method performs.

3. What type of keyword is used to change the access level of a method?

An access modifier.

4. What is another word used for describing the access level of a method?

Visibility.

5. Explain the scope of each of the variables in the code below.

var1: Local variable in the main method.

var2: Local variable inside the for loop in the main method.

var3: Local variable inside method1.

var4: Local variable inside the for loop in method1.

6. Write a method declaration for each description.

a)

java
public static int getVowels(String word)
b)

java
public static int extractDigit(int number)
c)

java
public static String insertString(String word, int number)
7. a) How does the compiler distinguish one method from another?

The compiler distinguishes methods by their method signatures, including the method name and parameter types.

b) Can two methods in the same class have the same name? Explain.

Yes. This is called method overloading. They must have different parameter lists.

8. a) What type of comments should be used for describing a method?

Documentation comments (Javadoc comments).

b) What three things should the comments for a method describe?

What the method does.

The parameters the method requires.

The value the method returns.

11. True or False
a) Breaking a task down into methods is called procedural abstraction.

True

b) A method call consists of the method declaration in an assignment statement.

False. A method call invokes a method to execute its instructions.

c) A void method must return a value.

False. A void method does not return a value.

d) An access modifier declares the return type of a method.

False. An access modifier determines who can access the method.

e) The keyword static declares a method is a class method.

True

f) Method parameters are enclosed by braces ( ).

False. Method parameters are enclosed by parentheses.

g) Local variables can be used by any method in a class.

False. Local variables can only be used within the method or block where they are declared.

h) The value of an argument passed to a method can be changed in an assignment statement in the method.

False. Changing a parameter does not change the original argument variable.

i) Method overloading means that an application contains more than 10 methods.

False. Method overloading means having multiple methods with the same name but different parameter lists.

j) The return statement is used to send a value back to the calling statement.

True

k) The precondition of a method states the data types of the method's parameters.

False. A precondition describes what must be true before a method runs.

l) The postcondition of a method describes the way the method accomplishes its task.

False. A postcondition describes what is true after the method finishes executing.